

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from google.colab import files

# Upload your dataset
uploaded = files.upload()

# Load dataset (replace with your filename)
df = pd.read_csv("DATA SET EXCEL.csv")

# Inspect dataset
print(df.head())
print(df.info())
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving DATA SET EXCEL.csv to DATA SET EXCEL.csv

	day	month	year	Temperature	RH	Ws	Rain	FFMC	DMC	DC	ISI	BUI	\
0	1	6	2012	29	57	18	0.0	65.7	3.4	7.6	1.3	3.4	
1	2	6	2012	29	61	13	1.3	64.4	4.1	7.6	1.0	3.9	
2	3	6	2012	26	82	22	13.1	47.1	2.5	7.1	0.3	2.7	
3	4	6	2012	25	89	13	2.5	28.6	1.3	6.9	0.0	1.7	
4	5	6	2012	27	77	16	0.0	64.8	3.0	14.2	1.2	3.9	

		FWI	Classes	Region
0	0.5	not fire	0	
1	0.4	not fire	0	
2	0.1	not fire	0	
3	0.0	not fire	0	
4	0.5	not fire	0	

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 243 entries, 0 to 242

Data columns (total 15 columns):

#	Column	Non-Null Count	Dtype
0	day	243 non-null	int64
1	month	243 non-null	int64
2	year	243 non-null	int64
3	Temperature	243 non-null	int64
4	RH	243 non-null	int64
5	Ws	243 non-null	int64
6	Rain	243 non-null	float64
7	FFMC	243 non-null	float64
8	DMC	243 non-null	float64
9	DC	243 non-null	float64
10	ISI	243 non-null	float64
11	BUI	243 non-null	float64
12	FWI	243 non-null	float64
13	Classes	243 non-null	object
14	Region	243 non-null	int64

dtypes: float64(7), int64(7), object(1)

memory usage: 28.6+ KB

```
# Check missing values
print(df.isnull().sum())
```

```
# Summary statistics
print(df.describe())
```

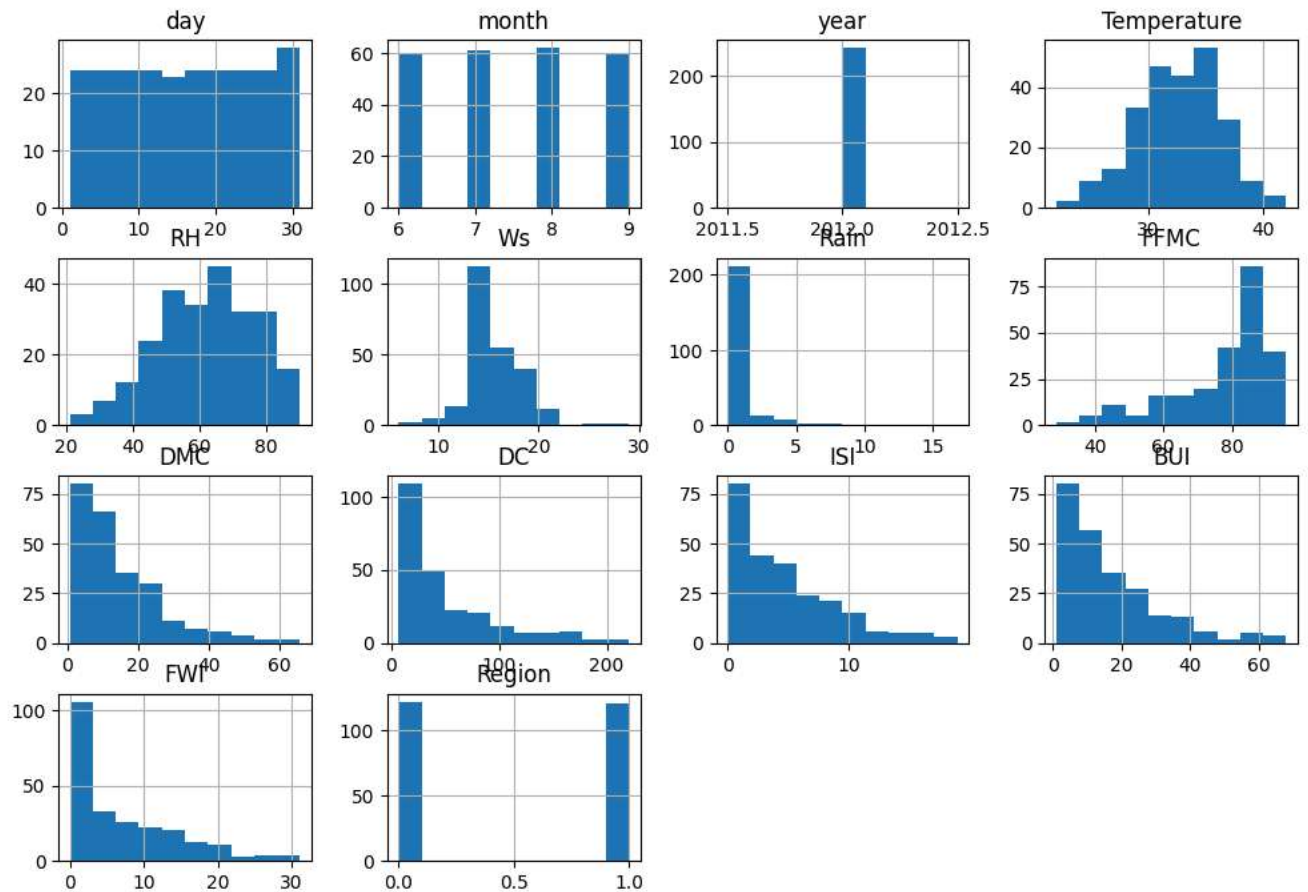
```
day          0
month        0
year         0
Temperature  0
RH           0
Ws           0
Rain         0
FFMC         0
DMC          0
DC           0
ISI          0
BUI          0
FWI          0
Classes      0
Region       0
dtype: int64
```

	day	month	year	Temperature	RH	Ws \
count	243.000000	243.000000	243.0	243.000000	243.000000	243.000000
mean	15.761317	7.502058	2012.0	32.152263	62.041152	15.493827
std	8.842552	1.114793	0.0	3.628039	14.828160	2.811385
min	1.000000	6.000000	2012.0	22.000000	21.000000	6.000000
25%	8.000000	7.000000	2012.0	30.000000	52.500000	14.000000
50%	16.000000	8.000000	2012.0	32.000000	63.000000	15.000000
75%	23.000000	8.000000	2012.0	35.000000	73.500000	17.000000
max	31.000000	9.000000	2012.0	42.000000	90.000000	29.000000

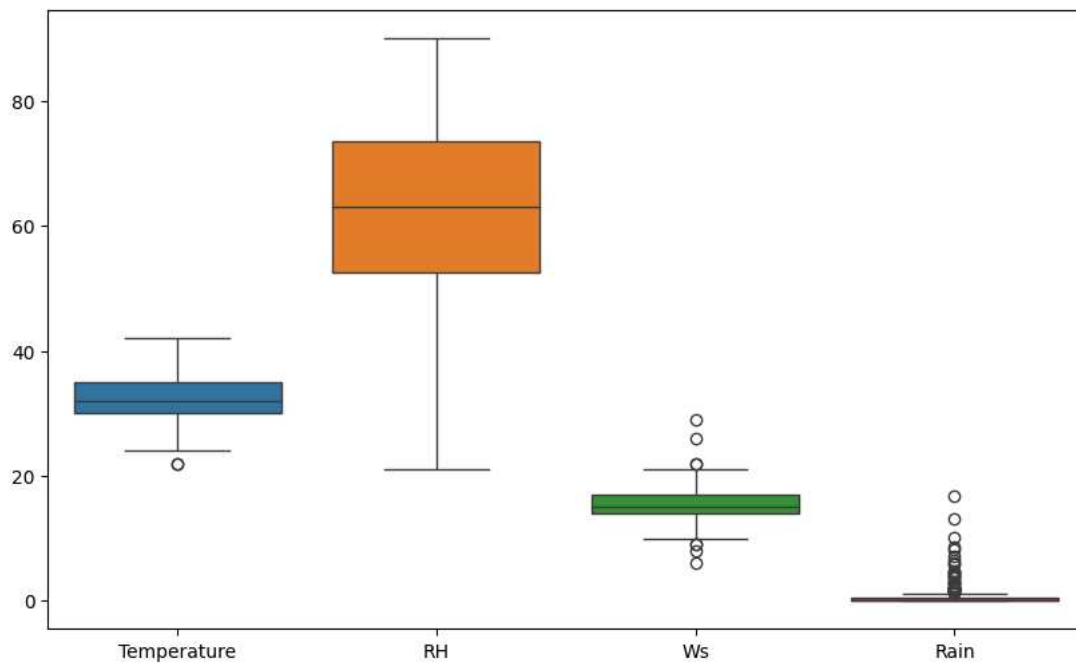
	Rain	FFMC	DMC	DC	ISI	BUI \
count	243.000000	243.000000	243.000000	243.000000	243.000000	243.000000
mean	0.762963	77.842387	14.680658	49.430864	4.742387	16.690535
std	2.003207	14.349641	12.393040	47.665606	4.154234	14.228421
min	0.000000	28.600000	0.700000	6.900000	0.000000	1.100000
25%	0.000000	71.850000	5.800000	12.350000	1.400000	6.000000
50%	0.000000	83.300000	11.300000	33.100000	3.500000	12.400000
75%	0.500000	88.300000	20.800000	69.100000	7.250000	22.650000
max	16.800000	96.000000	65.900000	220.400000	19.000000	68.000000

	FWI	Region
count	243.000000	243.000000
mean	7.035391	0.497942
std	7.440568	0.501028
min	0.000000	0.000000
25%	0.700000	0.000000
50%	4.200000	0.000000
75%	11.450000	1.000000
max	31.100000	1.000000

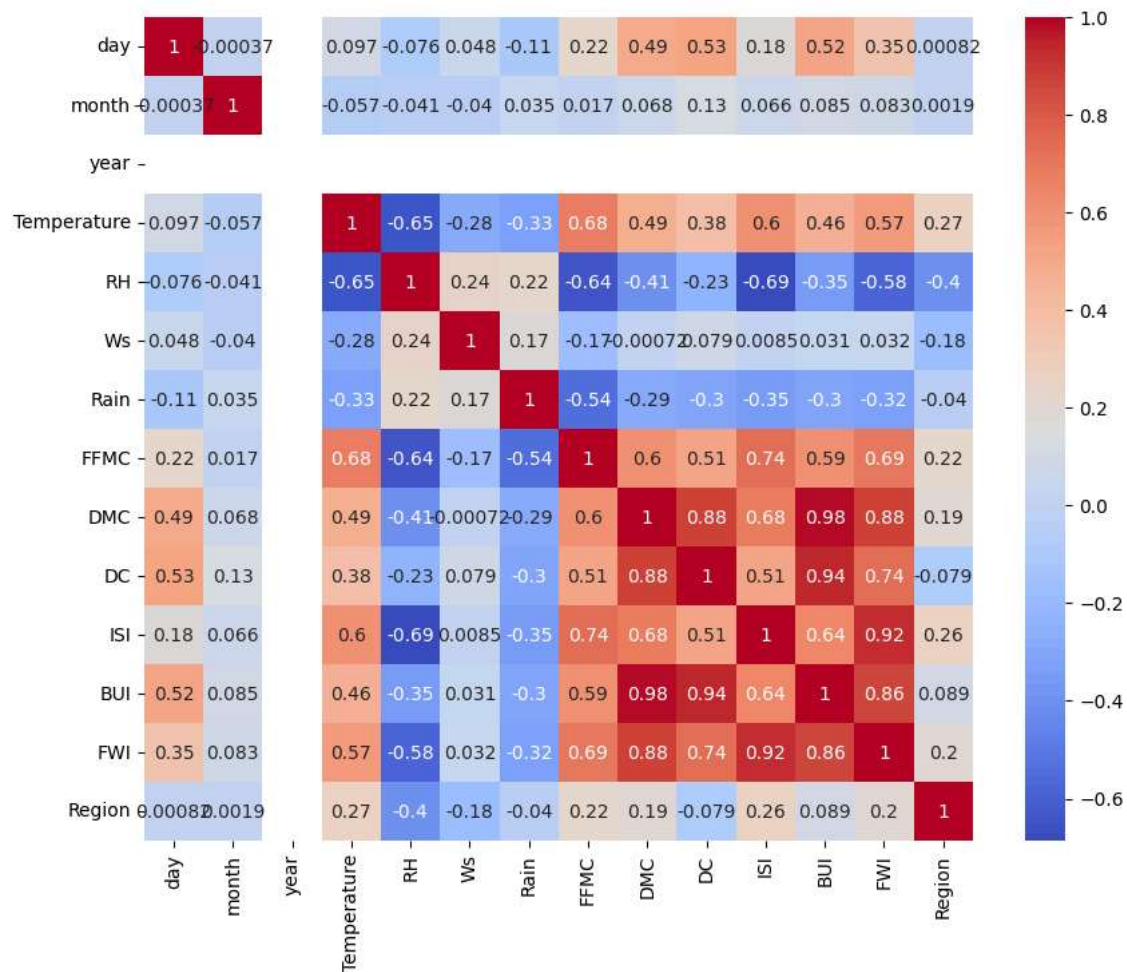
```
# Histograms for feature distributions
df.hist(figsize=(12,8))
plt.show()
```



```
# Boxplots for outlier detection
plt.figure(figsize=(10,6))
sns.boxplot(data=df[['Temperature','RH','Ws','Rain']]) # adjust columns if needed
plt.show()
```



```
# Correlation heatmap
plt.figure(figsize=(10,8))
sns.heatmap(df.select_dtypes(include=np.number).corr(), annot=True, cmap="coolwarm")
plt.show()
```



```
# Encode categorical variables (Classes, Region, Month, Day if needed)
df['Classes'] = df['Classes'].astype('category').cat.codes
df['Region'] = df['Region'].astype('category').cat.codes
```

```
df['month'] = df['month'].astype('category').cat.codes if 'month' in df.columns else df['month']  
df['day'] = df['day'].astype('category').cat.codes if 'day' in df.columns else df['day']
```

```
# Save cleaned dataset  
df.to_csv("cleaned_fwi_dataset.csv", index=False)  
print("Cleaned dataset saved successfully!")
```

```
# Download to your computer  
files.download("cleaned_fwi_dataset.csv")
```

Cleaned dataset saved successfully!